PILOTFISH · EDI 837 CLAIM SCRUB — DESIGN

Construction Replay Transcript

EDI 837 Claim Scrub — Design · 2026-08-13 · what you'll hear in documents/construction-replay.mp4

What this interface does

SQL Server professional claims to payer-profile edits (referring NPI / Place of Service) to kickout work queue or clean 837 + SNIP before the clearinghouse. Every scrub decision is copied to a BI outcome folder.

Who it's for

- Sources: Edi837ClaimScrub (Claims PENDING)
- Destinations: output/kickouts/, output/edi/, output/snip/, output/bi/
- Demo vs production: Synthetic payer rules mirrored in XSLT (not a live SQL join). SNIP on the clean path only.
- Anyone watching the construction video or reading this transcript

Systems involved

- Sources: Edi837ClaimScrub (Claims PENDING)
- Destinations: output/kickouts/, output/edi/, output/snip/, output/bi/
- Demo vs production: Synthetic payer rules mirrored in XSLT (not a live SQL join). SNIP on the clean path only.

Documents we also produce

EDI837_Claim_Scrub_Capability_Brief.pdf · ready
Short overview of what the interface does — shareable without reading route XML.

EDI837_Claim_Scrub_Test_Plan.pdf · ready
Scenarios that prove the interface works end-to-end.

Module documentation pack (12 deep-dive PDFs) · ready
PilotFish product PDFs for the modules used in this build.

EDI837_Claim_Scrub_V2_Route_Diagrams.pdf · ready
V2 route diagrams for visual review of module topology.

construction-replay.mp4 + this transcript · generate when ready
Narrated walkthrough of building the routes, then a live Demo-tab test when inject is available.

Narration

Spoken lines from the construction video (setup, build-replay, live test, closing).

Setup

Welcome

Welcome to the PilotFish demo: EDI 837 Claim Scrub. SQL Server professional claims to payer-profile edits (referring NPI / Place of Service) to kickout work queue or clean 837 + SNIP before the clearinghouse. PilotFish supports the whole flow out of the box.

Pipeline overview

Before we wire anything, here's the flow end to end. SQL Server professional claims to payer-profile edits (referring NPI / Place of Service) to kickout work queue or clean 837 + SNIP before the clearinghouse.

External systems

And here's what sits around the routes. SQL Server, PilotFish eiPlatform, and Demo Web UI. All of those are spun up in docker images for this demo.

1 - Poll SQL Server Claims

First route

SQL Server professional claims to payer-profile edits (referring NPI / Place of Service) to kickout work queue or clean 837 + SNIP before the clearinghouse. We'll build that in 2 routes. Starting with a blank canvas on the first route.

SQL step

First up is a SQL polling listener — it runs a query against Edi837ClaimScrub over JDBC. It checks about every 15 seconds.

Pretty-Print Raw Claims

This processor pretty-prints the XML — line breaks and indent — so the file on disk is readable.

Write Raw Claims Snapshot

Then we write that XML to disk so we can inspect it. That lands under /opt/pilotfish/output/claims.

Hand Off Claims

Here's a Route to Route processor — next step in the flow.

2 - Scrub Claim And Route

Next route

Next route — blank canvas again, then the modules one at a time.

Trigger listener

This is a programmable trigger listener. The first route hands the batch here so we can fork and process each row.

Fork - XPath

This XPath fork splits the file so each transaction set becomes its own message.

Read Claim Keys

XPath evaluation copies a few fields off the XML onto the transaction so later steps can use them.

Save Claim XML

The control file is just a name on one line. This processor copies that text onto the transaction so the later steps can use it without parsing the body again.

Pretty-Print Claim Snapshot

This processor pretty-prints the XML — line breaks and indent — so the file on disk is readable.

Debug Write Claim XML

This writes a debug copy of the XML so we can inspect what the route saw. That lands under /opt/pilotfish/output/claims.

Restore Claim XML For Evaluate

The control file is just a name on one line. This processor copies that text onto the transaction so the later steps can use it without parsing the body again.

Evaluate Payer Profile Edits

Now for the mapping — we're using the stock XSLT processor with transform-evaluate-payer-edits.xslt. It maps claimId, claimNumber, payerId.

Pretty-Print Scrub Decision

This processor pretty-prints the XML — line breaks and indent — so the file on disk is readable.

Read Scrub Disposition

XPath evaluation copies a few fields off the XML onto the transaction so later steps can use them.

Debug Write Decision XML

This writes a debug copy of the XML so we can inspect what the route saw. That lands under /opt/pilotfish/output/claims.

Write Kickout Work Queue File

Then we write that XML to disk so we can inspect it. That lands under /opt/pilotfish/output/kickouts.

Write BI Edit Outcome File

Then we write that XML to disk so we can inspect it. That lands under /opt/pilotfish/output/bi.

Restore Claim XML For Clean Path

The control file is just a name on one line. This processor copies that text onto the transaction so the later steps can use it without parsing the body again.

Map Claim To EDI XML

Now for the mapping — we're using the stock XSLT processor with transform-claim-to-edi-xml.xslt. Watch the for-each over lines. It maps payerId, ctrl, claimNumber.

Pretty-Print EDI XML

This processor pretty-prints the XML — line breaks and indent — so the file on disk is readable.

Write Intermediate EDI XML

Then we write that XML to disk so we can inspect it. That lands under /opt/pilotfish/output/claims.

Save EDI XML for SNIP

The control file is just a name on one line. This processor copies that text onto the transaction so the later steps can use it without parsing the body again.

XML to EDI 837

Here's the EDI transformation — it turns our EDI XML into X12 text on the wire.

Write 837 EDI File

Then we write the X12 text to disk. That lands under /opt/pilotfish/output/edi.

Restore EDI XML for SNIP

The control file is just a name on one line. This processor copies that text onto the transaction so the later steps can use it without parsing the body again.

EDI SNIP Validation (Types 1-3)

Here's an EDI SNIP Validation processor — next step in the flow.

Pretty-Print SNIP Results

This processor pretty-prints the XML — line breaks and indent — so the file on disk is readable.

Write SNIP Results File

Then we write that XML to disk so we can inspect it. That lands under /opt/pilotfish/output/snip.

No-Op Complete

This route has nothing to send outbound. The work was the trigger, so we end on a null transport.

Closing

Demo complete

That's the demo — sQL Server professional claims to payer-profile edits (referring NPI / Place of Service) to kickout work queue or clean 837 + SNIP before the clearinghouse. Thanks for choosing PilotFish.